

ATPL EASA STUDY GUIDE | 021 - AIRFRAME & SYSTEMS

# Chapter 15: AC Distribution, TRUs, Inverters & Architecture

Syllabus Source: CAE Oxford (pp. 139-166 (Elec)) | Pass Target: >= 90% AviationExam

1. BUSBAR ARCHITECTURES: SPLIT-BUS VS PARALLEL SYSTEMS

Commercial aircraft distribute electrical power through rigid copper or aluminum conductor rails called busbars (buses). Two distinct system philosophies govern AC generator connection:

| Distribution System         | Operating Philosophy & Generator Connection                                                                  | Aircraft Examples & Nuances                                                                                                                                                                                       |
|-----------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Split-Bus System (Isolated) | Generators NEVER operate in parallel!<br>Each AC generator powers its own independent AC busbar.             | Standard on twin-engine aircraft (A320, B737, B777, B787). If one generator fails, a Bus Tie Breaker (BTB) closes automatically so the remaining generator powers both sides. Paralleling is strictly prohibited! |
| Parallel System             | All AC generators are synchronized in frequency, voltage, and phase angle, connected to a common tie busbar. | Four-engine heavy transports (B747-400, older widebodies). Requires complex real (kW) and reactive (kVAR) load sharing circuitry in the GCU.                                                                      |

2. POWER CONVERSION UNITS: TRUS & STATIC INVERTERS

Modern airliners carry both AC and DC equipment, requiring two-way electrical power conversion:

- Transformer Rectifier Unit (TRU): Converts primary 115V AC (400 Hz) into 28V DC (nominal) to power DC busbars, charge batteries, and supply DC avionics. Contains a step-down transformer and a 6-phase bridge silicon diode rectifier. Has NO MOVING PARTS and requires cooling airflow.
- Static Inverter: Converts 28V DC (from aircraft batteries or DC buses) into 115V AC (single-phase, 400 Hz) to power essential flight instruments (standby attitude indicator, flight controls, navigation radios) during complete AC generator failure.

**[!] EXAM TRAP (AVIATIONEXAM) - TRU vs Inverter Conversion Direction Trap**

- TRU (Transformer Rectifier Unit): Converts AC -> DC (115V AC in -> 28V DC out)! -

Static Inverter: Converts DC -> AC (28V DC in -> 115V AC out)! - Mnemonic: Inverter INVERTS the normal flow (takes battery DC and produces AC)!

3. GENERATOR CONTROL UNIT (GCU) & AUTOMATIC LOAD SHEDDING

Each AC generator is supervised by an electronic Generator Control Unit (GCU) providing protection and switching:

- GCU Protective Trips: Monitors for Overvoltage, Undervoltage, Overfrequency, Underfrequency, Feeder Fault (differential current between neutral and bus), and Reverse Power. If a fault occurs, the GCU opens the Generator Breaker (GB) and de-excites the generator field.

- Automatic Load Shedding: If an electrical generator is lost in flight, the remaining generator may exceed 100% rated capacity. The electrical system automatically sheds non-essential utility loads (galley ovens, cabin entertainment, passenger seat power, secondary water heaters) to preserve electrical power for flight instruments and safety systems!

#### 4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

##### Q1: What is the operational function of a Transformer Rectifier Unit (TRU) in an aircraft electrical system?

[A] To convert 28V DC into 115V AC [B] To convert 115V AC into 28V DC [C] To regulate generator frequency at 400 Hz [D] To charge oxygen cylinders

###### [\*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. A TRU steps down 115V AC power and rectifies it through solid-state silicon diodes into 28V DC power for the aircraft's DC busbars.

##### Q2: In a twin-engine jet aircraft featuring a 'split-bus' electrical architecture, what prevents two AC generators from operating in parallel?

[A] A reverse current relay [B] Interlocked Bus Tie Breakers (BTBs) that ensure generators operate completely isolated and never in parallel [C] Static inverters [D] Thermal fuses

###### [\*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. In split-bus systems, parallel connection is prohibited. Control logic interlocks ensure that when an external generator or second engine generator connects, the bus tie opens or isolates.

##### Q3: Following an in-flight generator failure on a commercial airliner, what occurs automatically during 'load shedding'?

[A] Fuel is jettisoned from the wings [B] Non-essential electrical consumers (galley equipment, passenger entertainment) are disconnected automatically to prevent overloading the remaining generator [C] The APU shuts down [D] Flight controls switch to manual reversion

###### [\*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Load shedding disconnects high-draw commercial utility loads (galleys) to guarantee sufficient capacity for essential flight guidance, instruments, and pumps.